Roll No. | | | | | | | |

ODD END SEMESTER EXAMINATION DECEMBER- 2024

Name of the Course: **B.Tech** Semester: **7**
Name of the Paper: **ASIC Design and FPGA** Paper Code: **TEC 752**

Time: 3 Hours Maximum Marks: 100

Note:-
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	(20marks)	
(a)	**With the help of suitable flowchart, explain how ASIC and FPGA are designed?**	CO1
(b)	**Describe the working principle of a CMOS inverter. Draw the voltage transfer characteristic and explain the significance of noise margins in a CMOS inverter.**	
(c)	**Give the comparison between VHDL and Verilog HDL with the help of examples.**	
Q2	(20 marks)	CO2, CO3
(a)	**With reference to ASIC and FPGA Families, explain Static RAM, EPROM and EEPROM technology.**	
(b)	**Write the Verilog or VHDL description of 4-bit Comparator.**	
(c)	**Explain the Anti-fuse technology with few examples.**	
Q3	(20 marks)	CO4
(a)	**What do you mean by Logic Synthesis? Discuss the role of Logic Synthesis in FPGA design cycle.**	
(b)	**What is meant by UCF in Xilinx? Explain its uses.**	
(c)	**Discuss Xilinx and Altera FPGA family.**	

Q4	(20 marks)	
(a)	**With reference to FPGA, discuss the concept of partitioning.**	
(b)	**Explain Floor Planning and Placement in FPGA implementation.**	CO5
(c)	**Discuss the various Routing Algorithm in FPGA implementation flow.**	
Q5	(20 marks)	
(a)	**Discuss the basic concept of Bus based communication architectures.**	CO6
(b)	**Discuss the FPGA design flow using Xilinx.**	
(c)	**Write a short note on System on Chip.**	